

2402-COL719

Project: 3D-memory(TSV)
Milestone 3

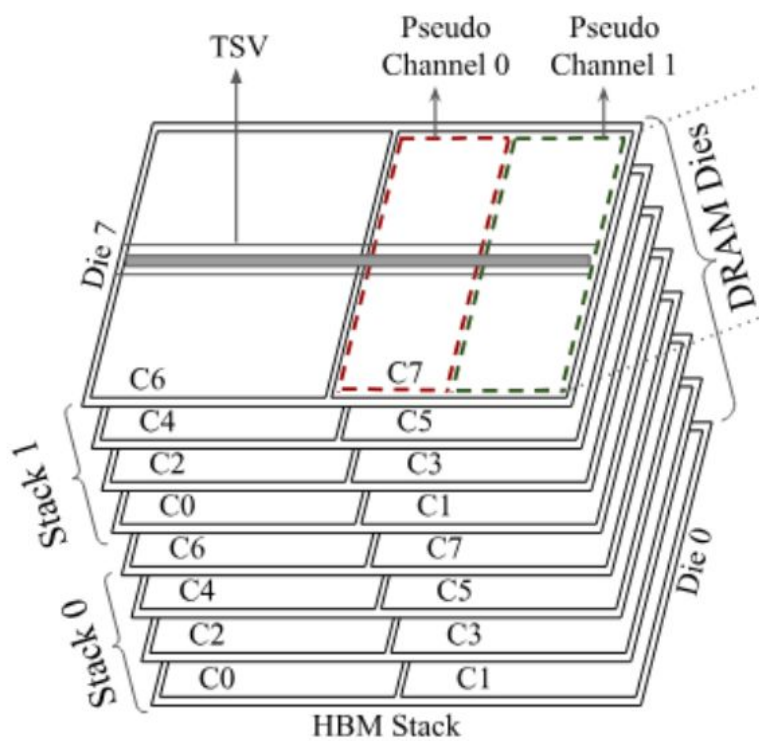
Submission of

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Objective

- Understand HBM working
- Experiment with different synthetic workloads utilizing 16 cores and 8 channels HBM
- Design a suitable Dynamic Thermal Management(DTM) policy.

High Bandwidth Memory(HBM) with Through Silicon Via(TSV)



- 4 X 4 X 8 banks
- 16 banks per channel
- Horizontally placed banks/ranks form channel.
- Channels at the bottom 4 layers are duplicated at the top 4 layers
- 8 channels (16 Pseudochannels)
- Each channel is independent of other.
- Various memory interleaving policies to map physical address to banks/ranks in and across channels
- Various access scheduling policies and channel partitioning policies

Simulation description

- Sampling interval : 1ms
- Temperature gradient is solved using Hotspot tool, which is provided a power trace file, layer configuration file that includes floorplan files for each layer, an initial temperature file and a configuration file for heatsink and leakage power model.
- Grid points are the centers of each banks.
- Power trace is calculated using model involving constant power per read/write access and constant refresh power.
- The experiment involve trying different read-write access rates sequence, that approximates various testbenches, and different Dynamic Thermal Management Policies.

Workloads description

- Random sequence of numbers as rates of memory access trace where :
 - Memory-intensive program have access rates in range 10,000/s to 11,000/s.
 - Memory-lightweight programs have most access rates in range 0 to 2,000/s while some(5%) in 7,000/s to 10,000/s.
 - Mixed intensity programs have half of the numbers in 0 to 4,000/s and rest in 4,000/s to 8,000/s throughout simulation.
- Each core is allocated separate Pseudo-channels of HBM, to reduce memory interference. ¹
- The Workloads comprises:
 - Memory-intensive programs running on all cores throughout the simulation.
 - Memory-intensive programs running on 8 cores, and mixed-intensity on other 8 cores.
 - Memory-intensive on 4 cores, Mixed-intensity on 8 cores, Light-intensity on 4 cores.
 - Mixed-intensity on 8 cores and Light-Intensity on 8 cores.
 - Light-intensity on all (16 cores)

- For each workload, longest runtime program and the core on which it runs (and thus the pseudochannel allocated to it) are determined probabilistically (uniform distributions).
- Each DTM is compared based on the actual time of completion of the above program, as well as, the peak temperature throughout time.

Dynamic Thermal Management (DTM) Policy - 1

- A primitive DTM is tested to be effective enough to just keep the peak temperatures below thresholds.
- The Policy solely involves putting Hot-channels (i.e. peak temp. of any banks in the channel exceeds threshold) to Low-Power mode(LPM) that involves preventing accesses (and thus dynamic power dissipation).
- Policy description:
 - Channels with peak temp. exceeding T_0 is put to LPM. If the channel cools down to $T_{0_c}(<T_0)$, it is put in active state.
 - $T_{0_c} < T_0$ $T_0 = 81$, $T_{0_c}=79$
- To make it application aware, cores running memory-intensive programs are allocated pseudochannels closer to the heat sink. This can be achieved with the help of the system software page mapper.
- Drawback: More than half of the cores are stalled most of the time .

Dynamic Thermal Management (DTM) Policy - 2

- Another DTM is also tested to be effective enough to just keep the peak temperatures below thresholds.
- The Policy solely involves putting Hot-channels (i.e. peak temp. of any banks in the channel exceeds threshold) to Low-Power mode(LPM) that involves preventing accesses (and thus dynamic power dissipation).
- Policy description:
 - Channels with peak temp. exceeding T_0 is put to LPM. If the channel cools down to $T_{0_c} (< T_0)$, it is put in active state.
 - If channel's peak temp. further exceeds $T_1 (> T_0)$, another neighbouring channel in contact with the highest temp. is put to LPM and brought back to active state once peak temp. goes below $T_{1_c} (< T_1 \text{ and } > T_0)$.
 - Another neighbouring channel is put to LPM if the channel's peak temp further exceeds $T_2 (> T_1)$ and brought back to active state once peak temp. goes below $T_{2_c} (< T_2 \text{ and } > T_1)$.
 - This continues until all 4 closest neighbours are put to LPM beyond T_4 .
 - $T_{0_c} < T_0 < T_{1_c} < T_1 < T_{2_c} < T_2 < T_{3_c} < T_3 < T_{4_c} < T_4$
- $T_0 = 76, T_1 = 77, T_2 = 78, T_3 = 80, T_4 = 81$
- $T_{0_c} = 74, T_{1_c} = 76.5, T_{2_c} = 77.5, T_{3_c} = 79, T_{4_c} = 80.5$

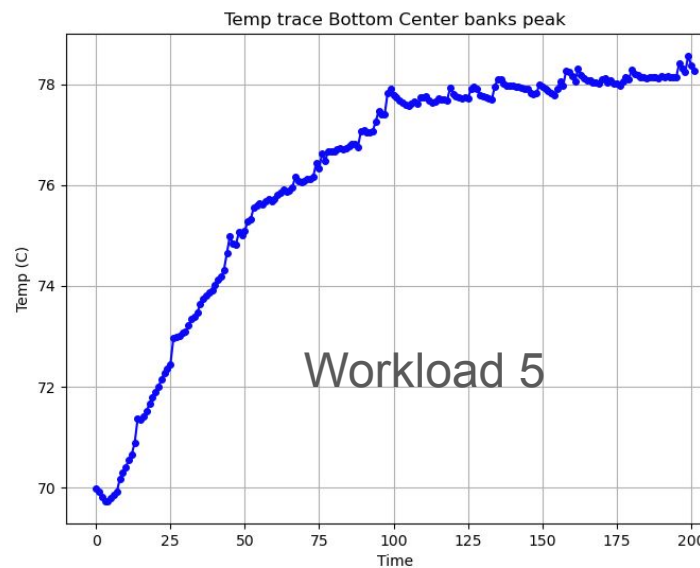
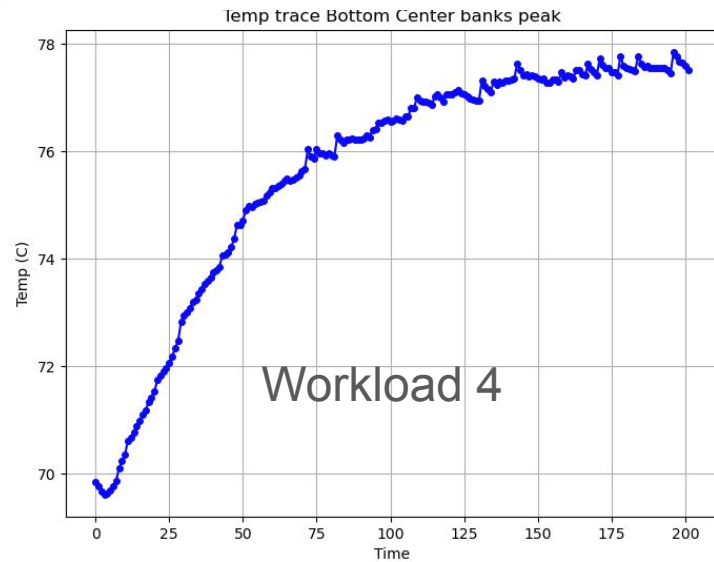
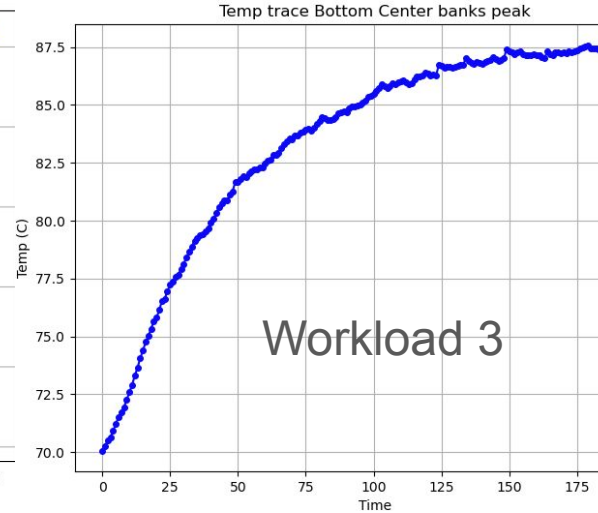
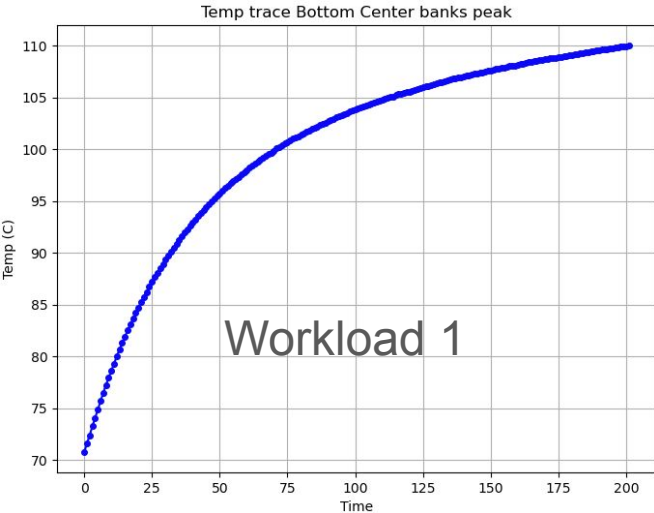
- The same System-level page mapper software can be used to map high-intensity program pages to pseudochannels closer to the heat sink.
- Drawback of DTM: More than half of the cores are stalled most of the time .

Dynamic Thermal Management (DTM) Policy - 3

- Similar to DTM-1 except thresholds for each layers of the HBM are different.
- $T_0 = [77.5, 78, 78.5, 79, 79.5, 80, 80.5, 81]$ thresholds for each layer
- $T_{0_c} = [75, 75.5, 76, 77, 78, 78.5, 79.5, 80.5]$
- Drawback of DTM: Still More than half of the cores are stalled most of the time .

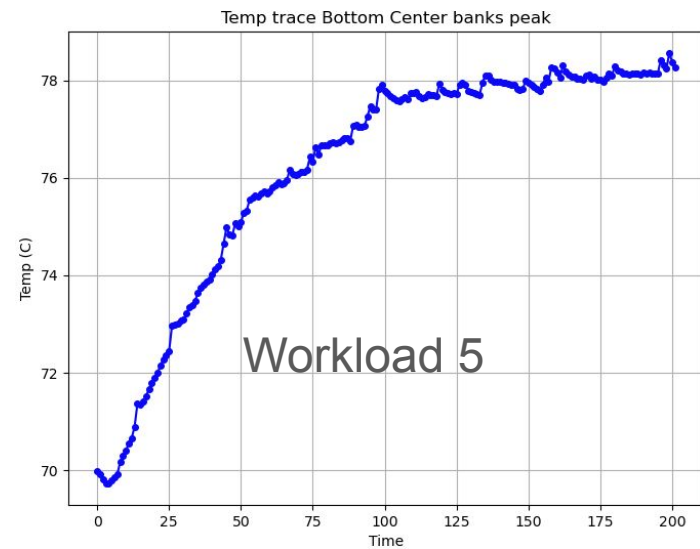
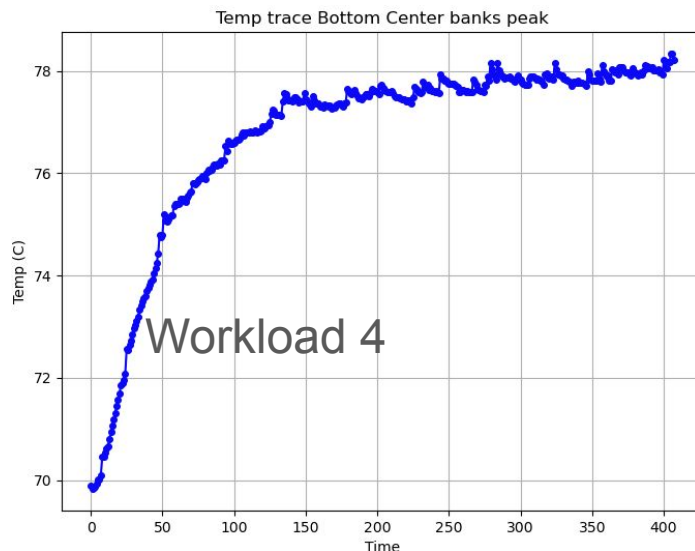
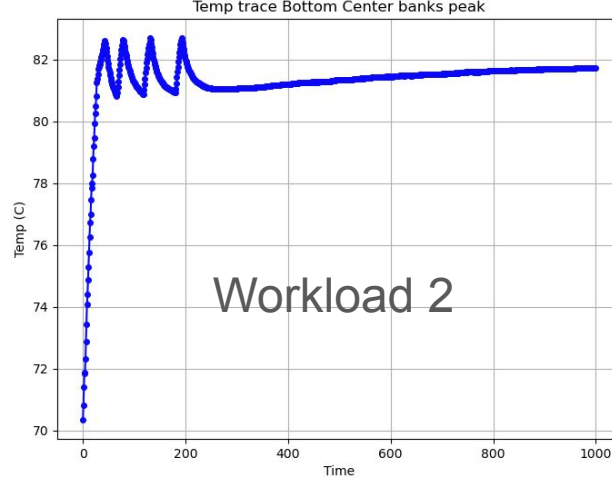
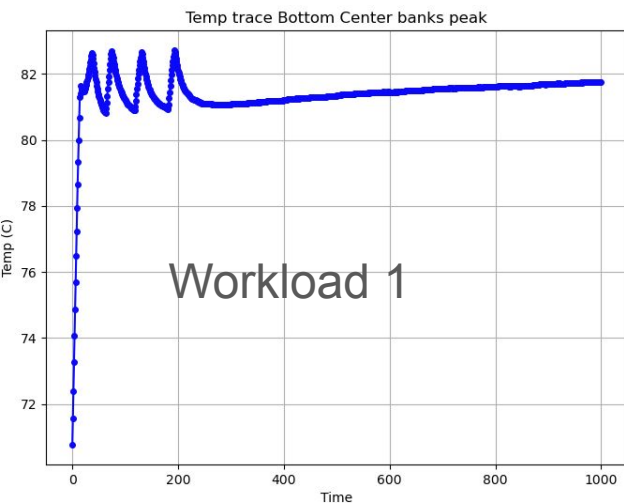
Dynamic Thermal Management (DTM) Policy - 4

- This Policy involves hybrid approach to the thermal management, by either putting the hot-channel to LPM or swapping hot-channel with another cooler channel or both.
- For swapping, it is expected that destination channel should have higher cooling rate or lower heating rate than source.
- Access-pattern-aware swapping, where swapping to another channel is preferred when the ratio of dynamic power to static power is high while putting to LPM is preferred when it is low.
- Policy description:
 - Swapping overhead is modelled by assuming a clock speed of memory to be 2 GHz (max 3.2 GHz)
 - Hotchannel at $\text{Temp} > T_0$ is swapped with cooler channel with $T < T_{0_c}$ where $T_0 > T_{0_c}$ ($T_0 = 78$ and $T_{0_c} = 76$), if the cooler channel has light(<5000) access rate, and is closer to heat sink.
 - Else, as channel temp $> T_1$, it is put into LPM until channel temp $< T_{1_c}$.
- This is the second best policy among others



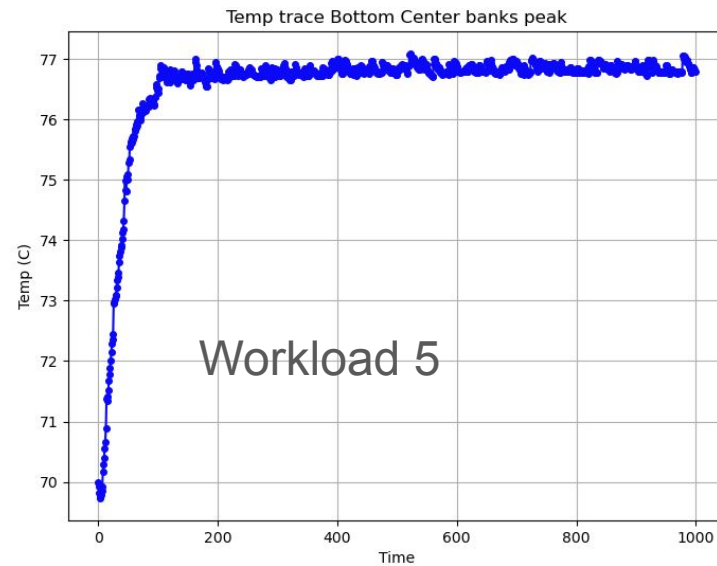
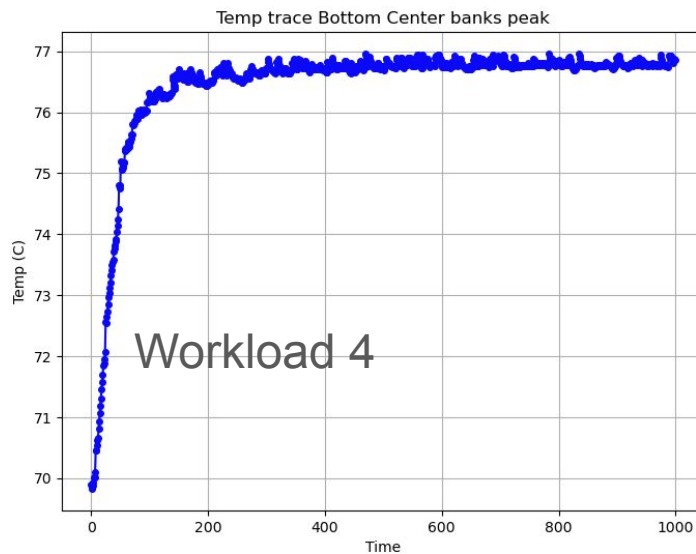
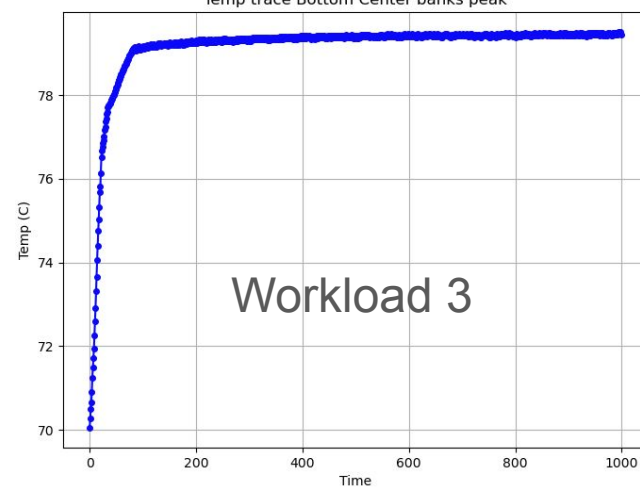
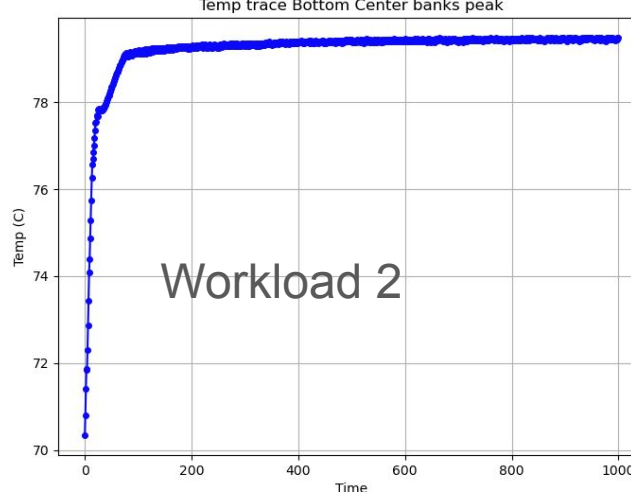
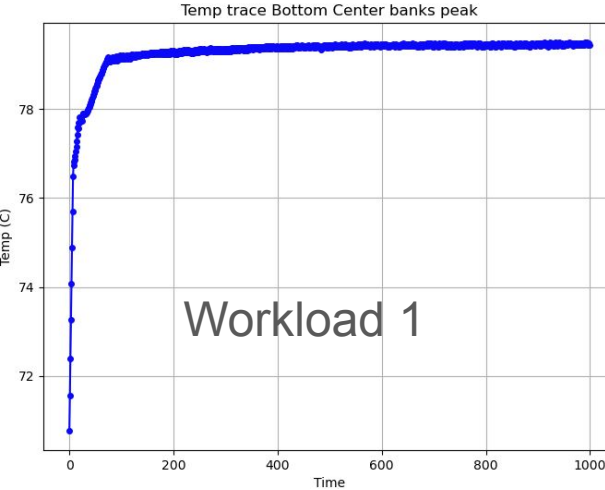
No DTM

Bottom
face center
banks



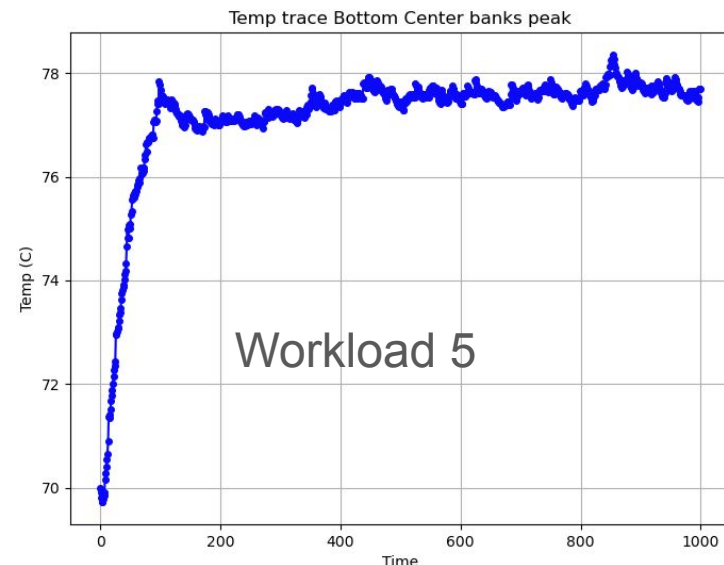
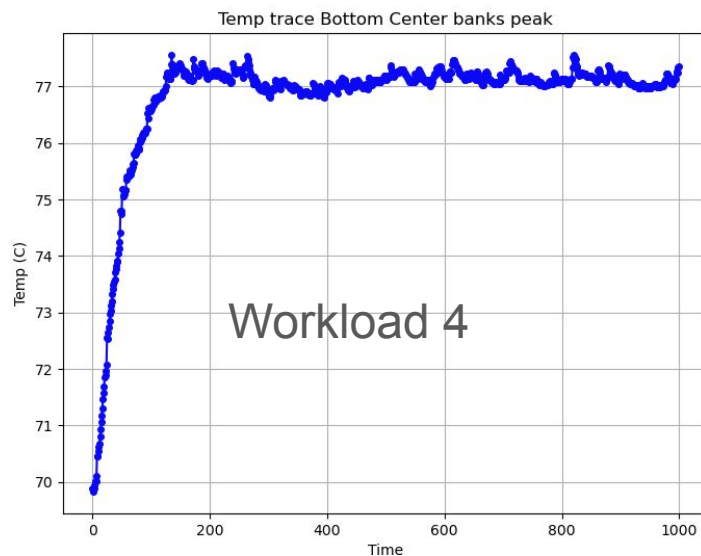
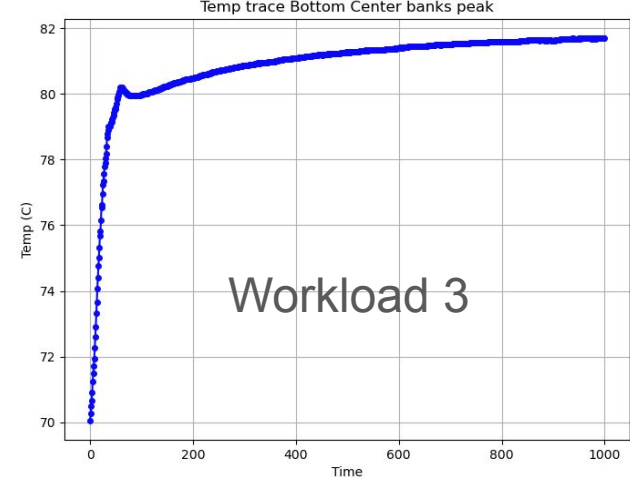
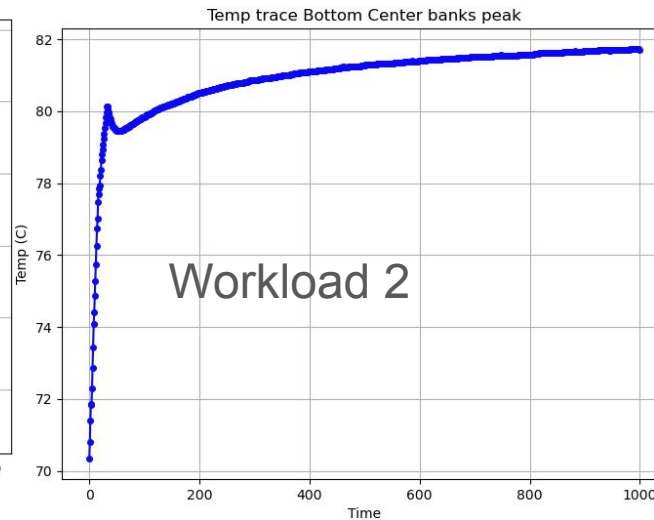
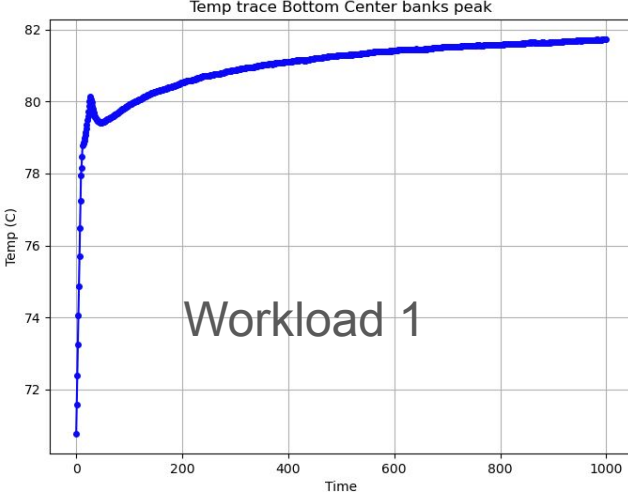
With
DTM-1

Bottom
face center
banks



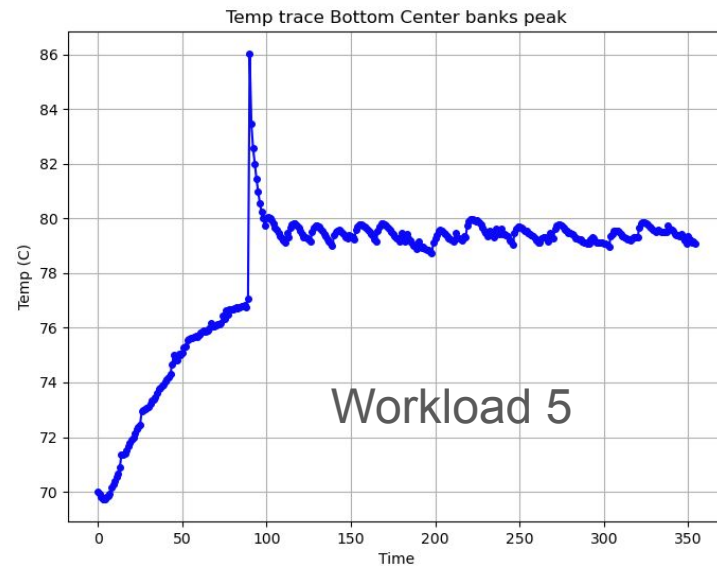
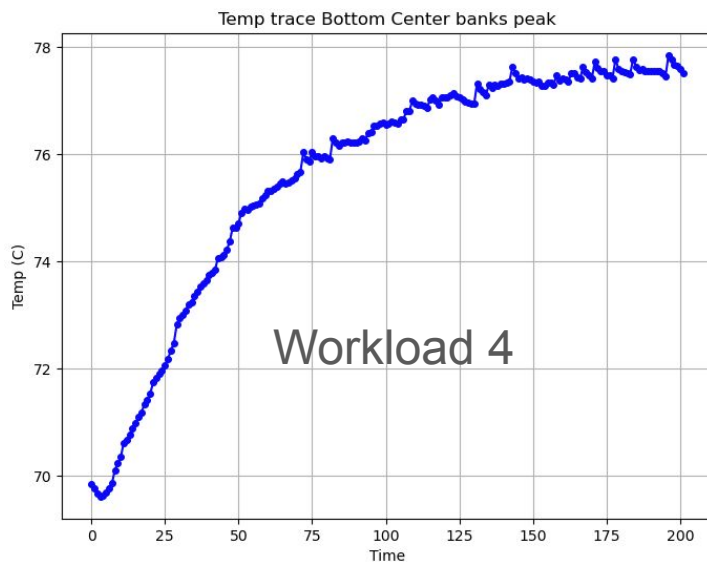
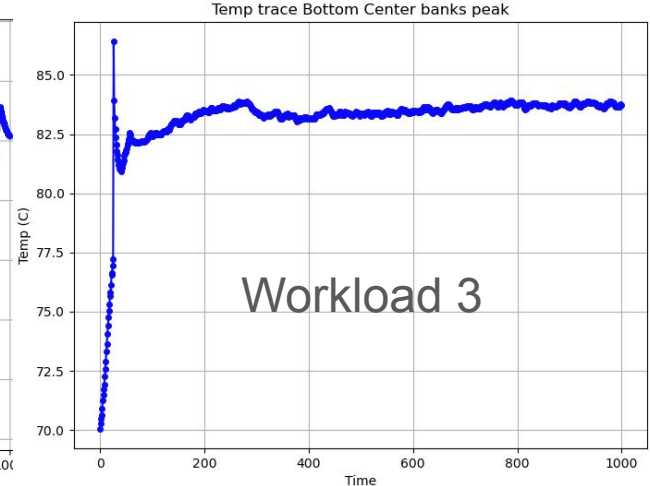
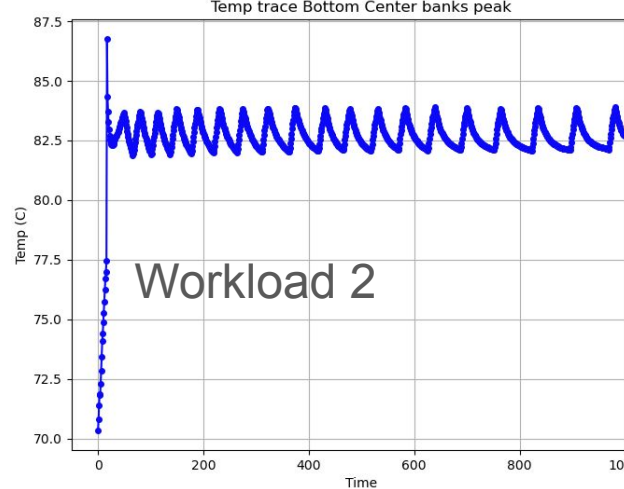
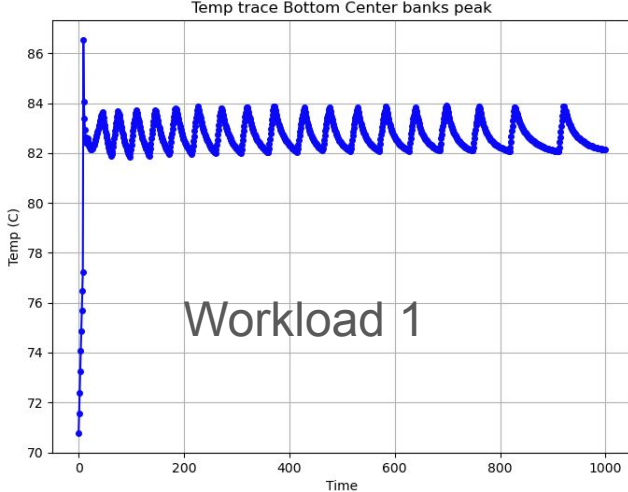
With
DTM-2

Bottom
face center
banks



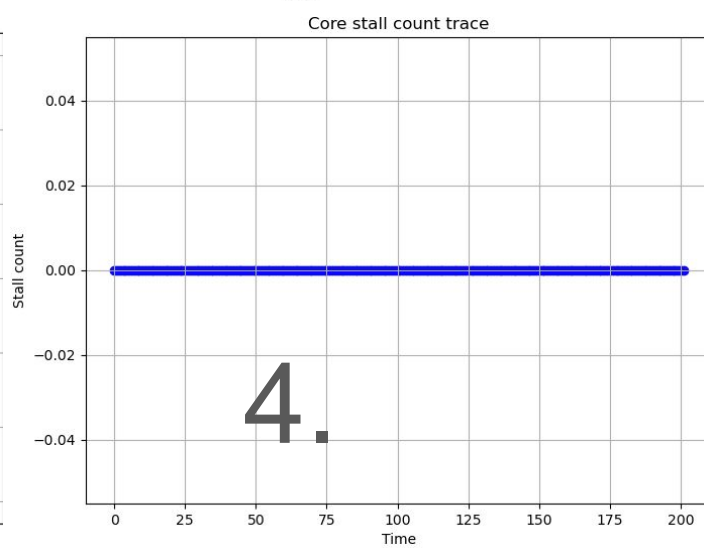
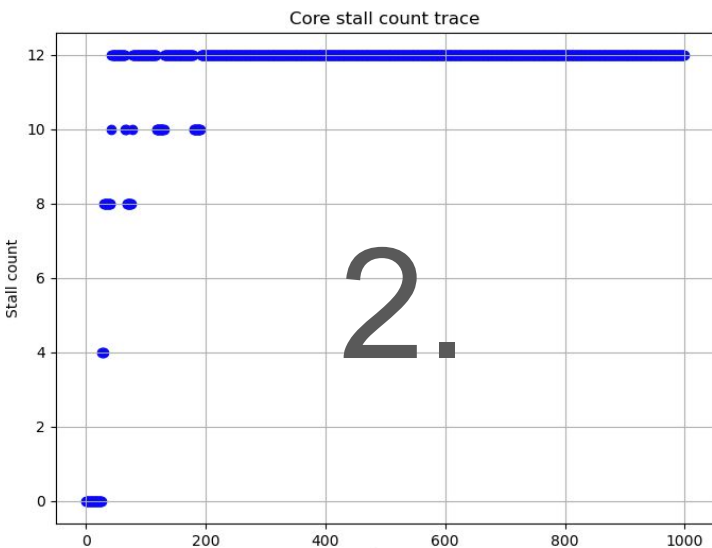
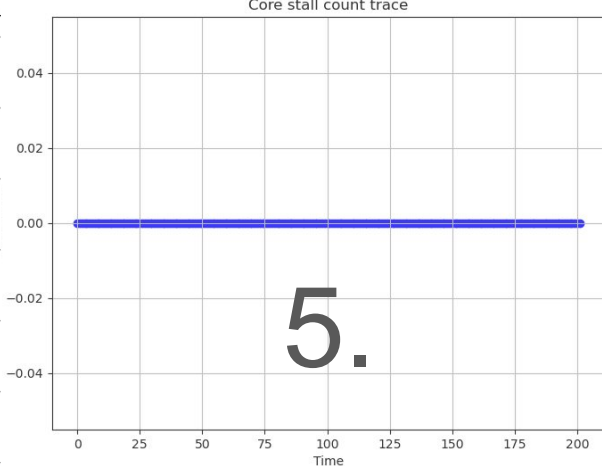
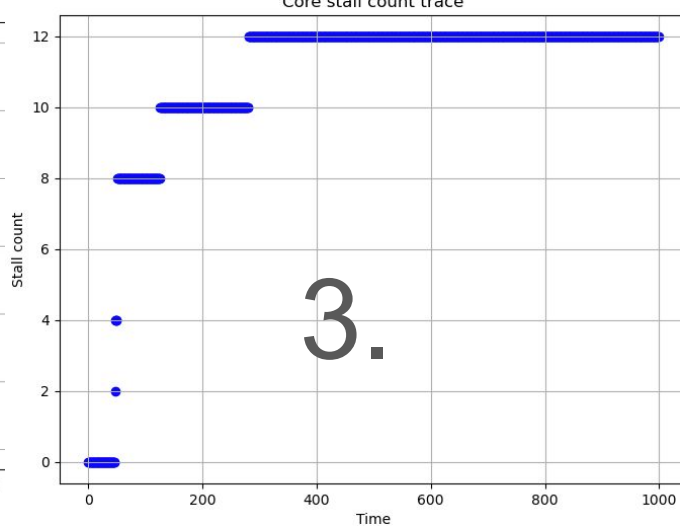
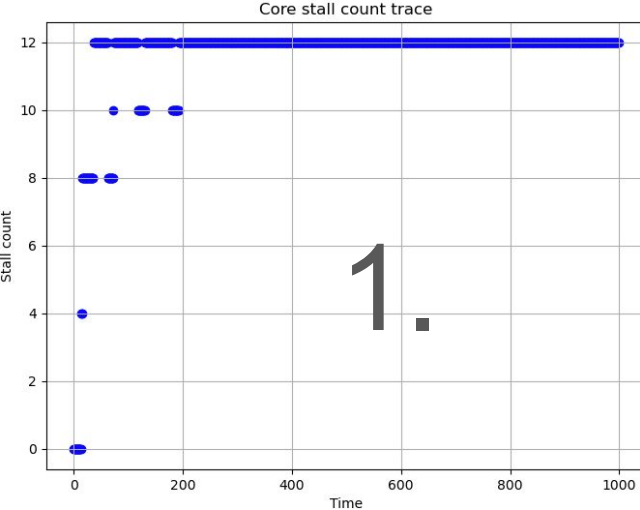
With
DTM-3

Bottom
face center
banks

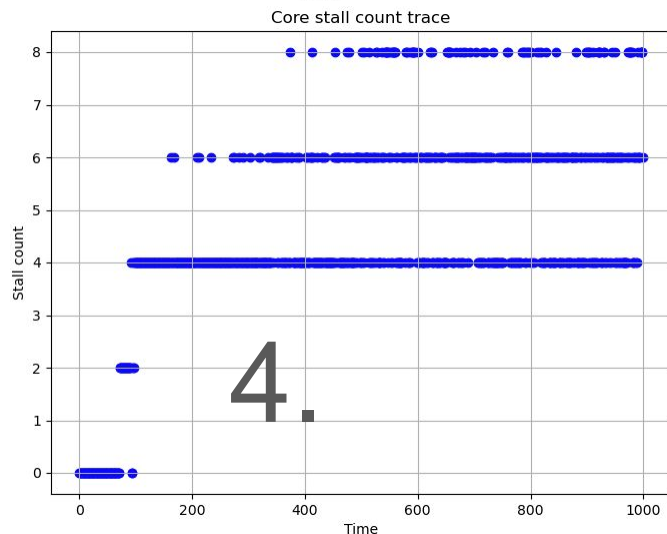
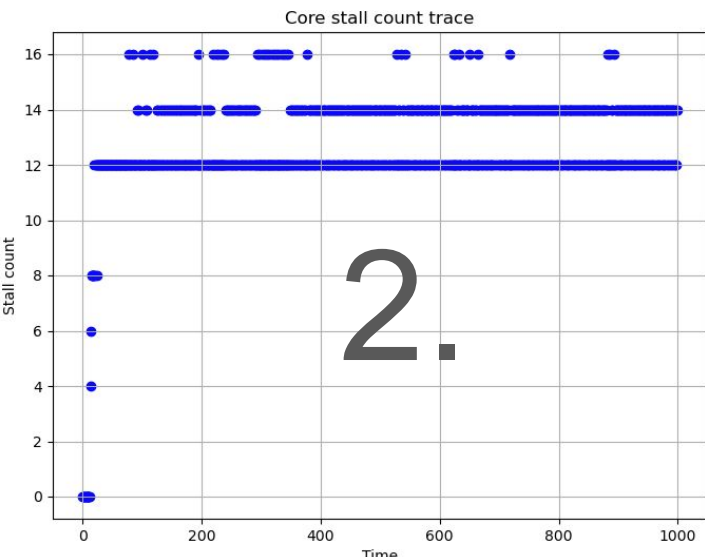
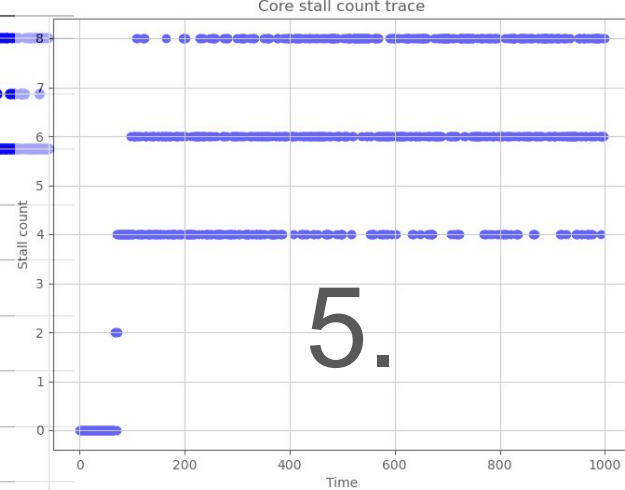
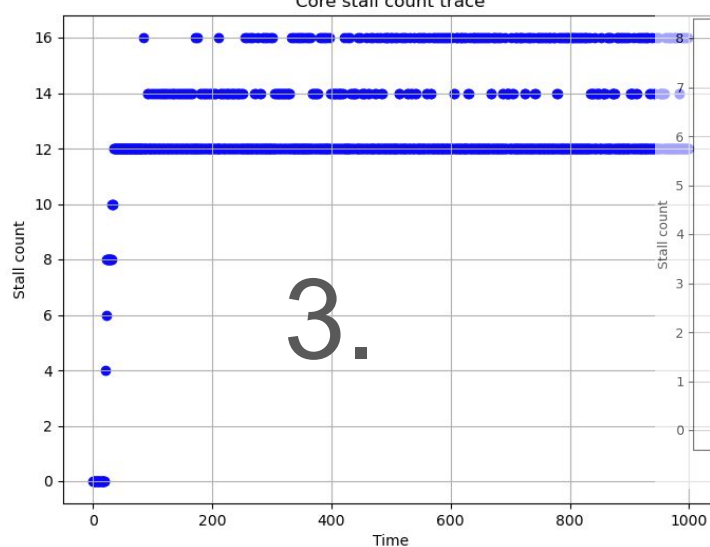
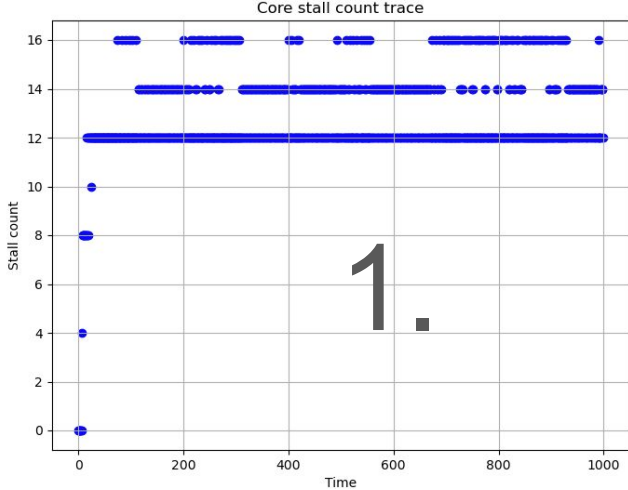


With
DTM-4

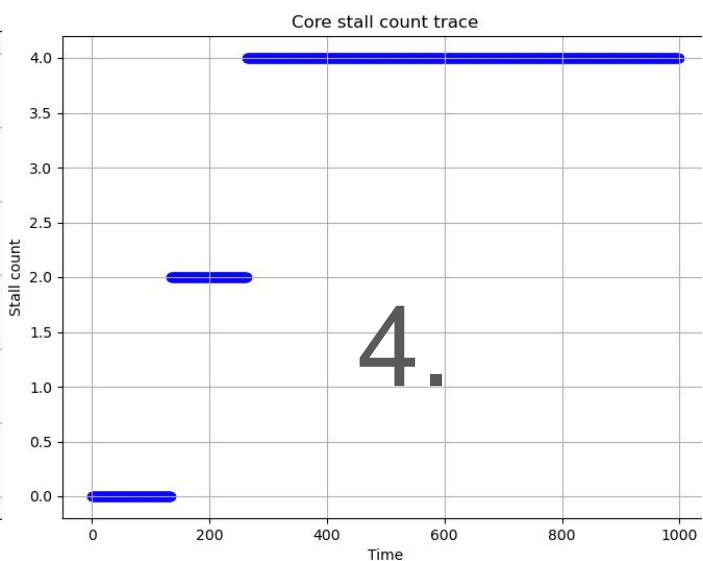
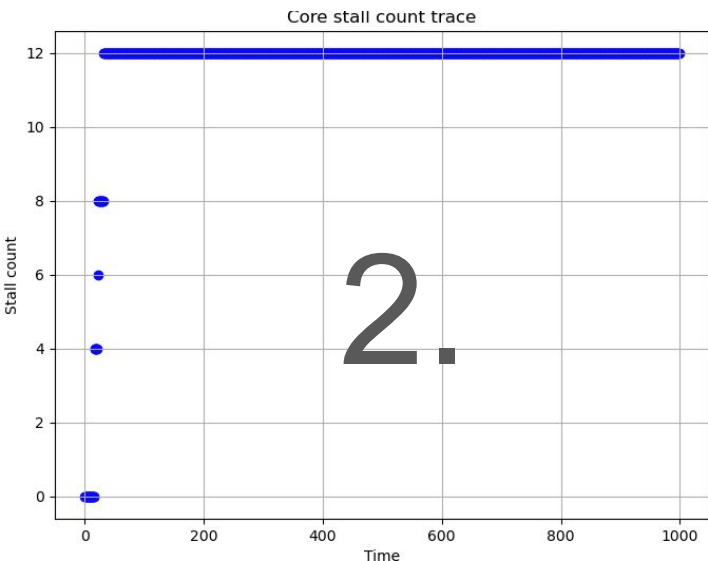
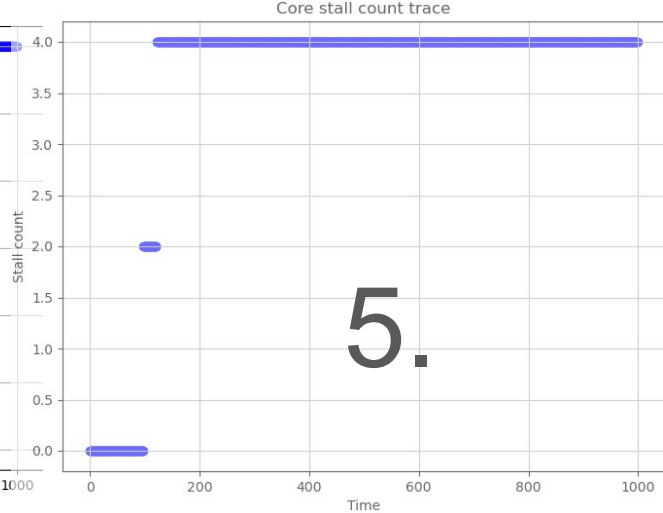
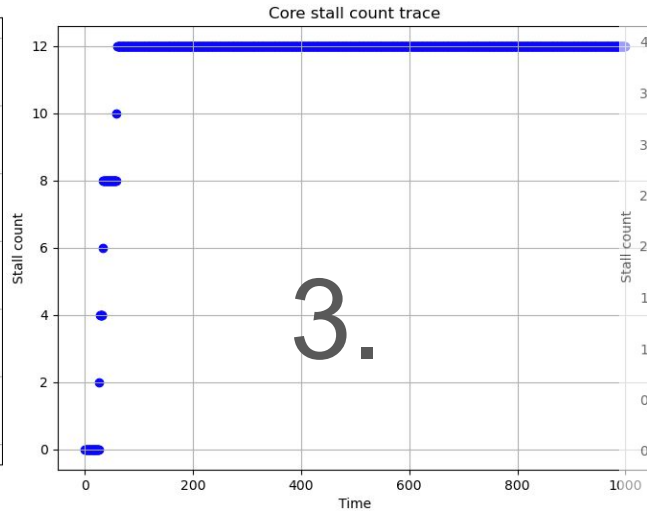
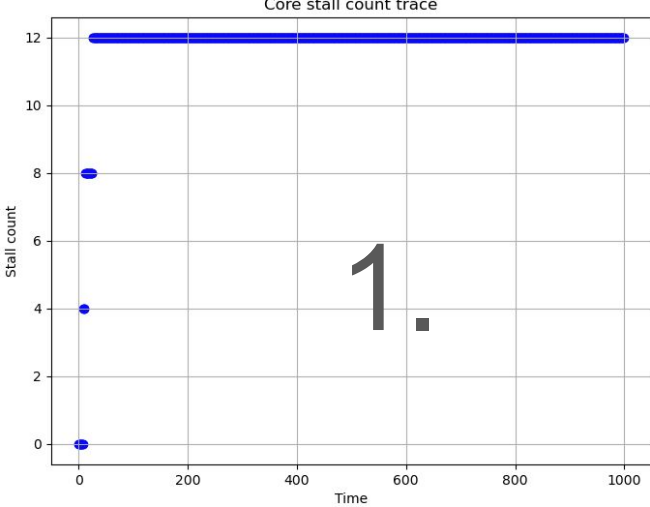
Bottom
face center
banks



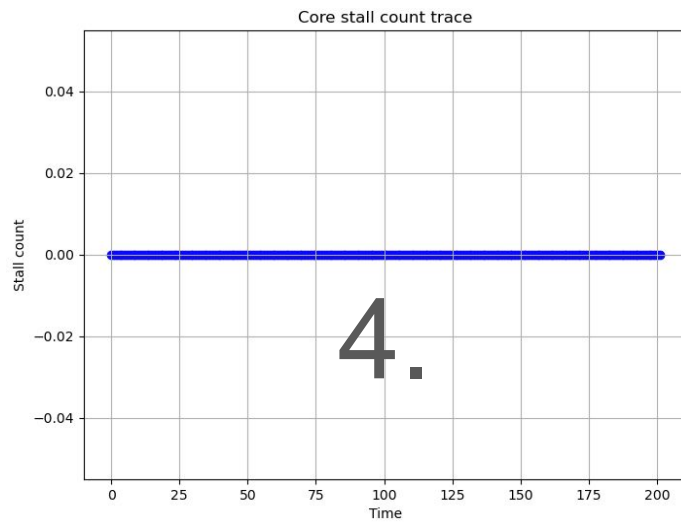
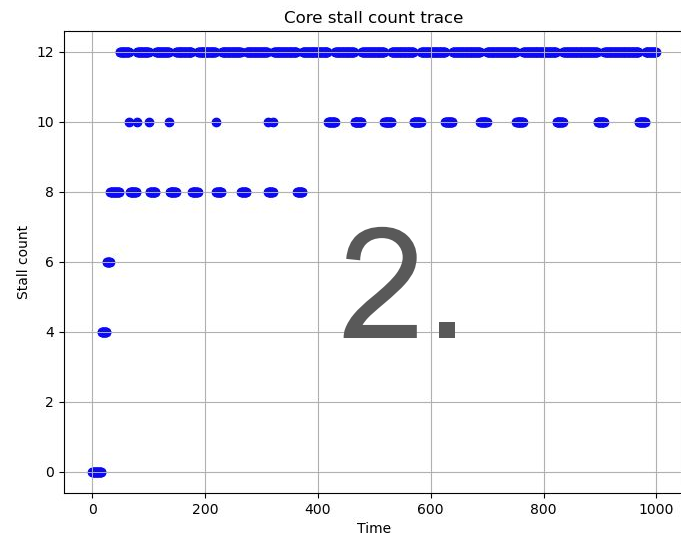
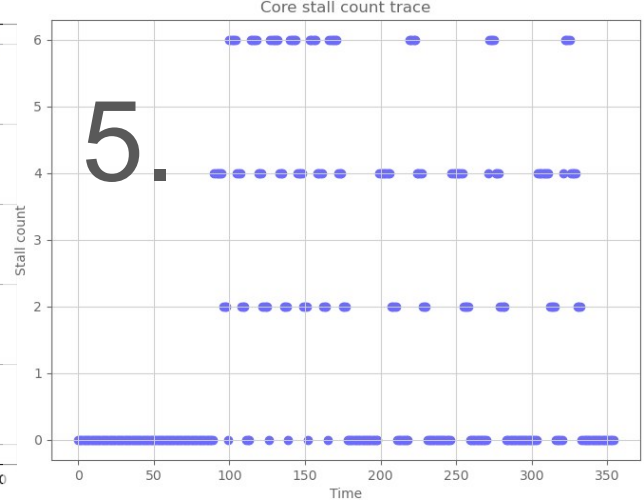
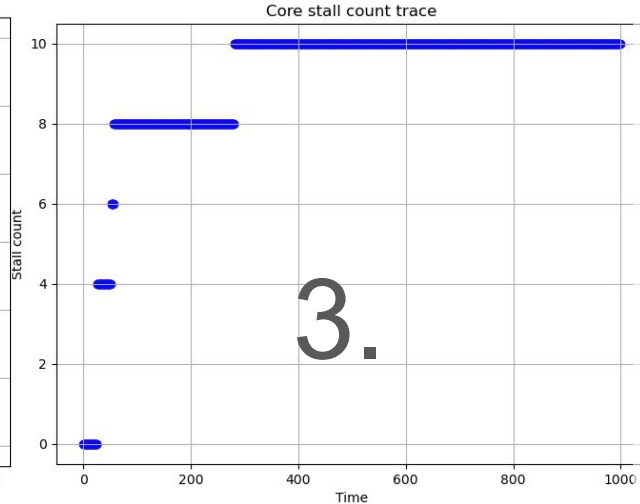
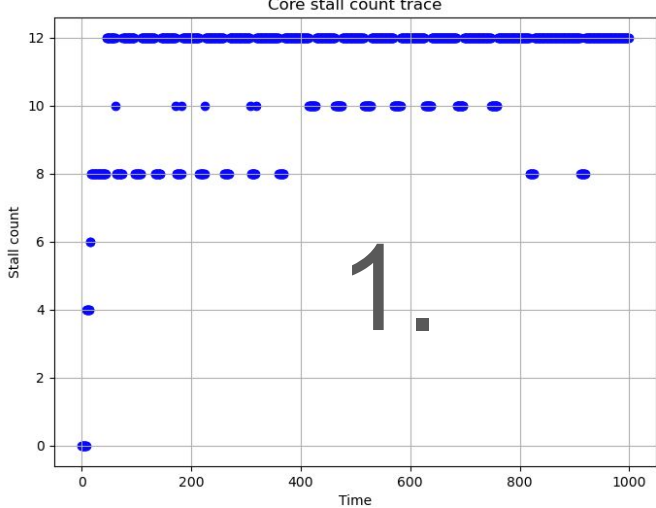
Stalled cores
over time-
DTM - 1



Stalled cores
over time-
DTM - 2



Stalled cores
over time-
DTM - 3



Stalled cores
over time-
DTM - 4

Longest program runtime comparison

DTM\ WORKLOADS	Workload 1	Workload 2	Workload 3	Workload 4	Workload 5
No DTM	201 ms	201 ms	201 ms	201 ms	201 ms
DTM 1	1347 ms	1136 ms	1034 ms	201 ms	201 ms
DTM 2	1564 ms	1783 ms	1527 ms	1128 ms	1132 ms
DTM 3	1457 ms	1378 ms	1307 ms	1034 ms	1079 ms
DTM 4	1193 ms	1185 ms	1176 ms	201 ms	354 ms

Observations

- DTM 1 and DTM 4 show good results compared to others. DTM1 is comparatively better in both metrics (max runtime program and peak temp over time) than DTM 4. The Swapping overhead also cause large temp rise.

Further Work

- Design of more realistic workloads, and can try using scheduling policy on top of the channel allocation.
- Idea of layerwise thresholds can be explored further, optimizing the layer thresholds based on gained insights.
- We might need to include some more workloads, because current testbench is not analyzed yet if it covers most workload instances, like kinds of low access workloads

Notes

- **The code include many dtm functions, like dtm4() is for DTM-1, dtm3() is for DTM-2, dtm6() is for DTM-3, and dtm5() is for DTM-4 of this report**

For all of the plots, visit here :

https://github.com/heysayan/COL719_assignment_comet/tree/proj719/hotspot_tool/visualization